

## BIGFORK, MN, USA

WMO: 727468

Lat: 47.783N

Lon: 93.650W

Elev: 1349

StdP: 13.99

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 04928

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -25.6      | -18.1 | -32.6                         | 0.9 | -26.3 | -25.9 | 1.4 | -18.2 | 19.0                  | 21.1 | 17.1 | 21.0 | 0.8                   | 290  | 0.575 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 24.0                   | 84.5            | 70.5 | 82.1 | 68.5 | 80.8 | 67.3 | 74.1                | 81.5 | 71.7 | 79.4 | 69.4 | 77.2 | 9.2                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 72.1                            | 124.8 | 79.1 | 69.8 | 115.2 | 77.3 | 66.2 | 101.6 | 74.0 | 38.5          | 82.4 | 36.4 | 79.2 | 34.4 | 77.5 | 83.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 17.3          | 15.4 | 12.8 | DB                         | -32.4 | 90.1               | 7.5 | 3.0                                                | -37.8 | 92.2       | -42.2 | 94.0       | -46.4 | 95.7       | -51.8 | 97.9 | (4) |
| (5)               |      |      | WB                         | -32.3 | 76.5               | 6.7 | 2.5                                                | -37.2 | 78.3       | -41.1 | 79.8       | -44.9 | 81.2       | -49.8 | 83.0 | (5) |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 39.4   | 8.4   | 11.2  | 25.9  | 40.2  | 52.1  | 61.2  | 66.2  | 63.3  | 56.0  | 43.3  | 29.1  | 13.9  |
|                                                                                  | DBStd       | 22.79  | 14.33 | 13.49 | 13.93 | 9.74  | 8.46  | 6.62  | 5.86  | 6.27  | 8.44  | 8.96  | 11.36 | 13.55 |
|                                                                                  | HDD50       | 5557   | 1290  | 1087  | 750   | 316   | 75    | 3     | 0     | 1     | 34    | 251   | 631   | 1119  |
|                                                                                  | HDD65       | 9568   | 1755  | 1507  | 1212  | 744   | 410   | 149   | 56    | 107   | 291   | 675   | 1078  | 1584  |
|                                                                                  | CDD50       | 1681   | 0     | 0     | 3     | 23    | 139   | 339   | 504   | 415   | 213   | 42    | 3     | 0     |
|                                                                                  | CDD65       | 219    | 0     | 0     | 0     | 1     | 10    | 36    | 95    | 56    | 20    | 1     | 0     | 0     |
|                                                                                  | CDH74       | 2583   | 0     | 0     | 1     | 12    | 163   | 434   | 1106  | 652   | 197   | 18    | 0     | 0     |
|                                                                                  | CDH80       | 574    | 0     | 0     | 0     | 2     | 31    | 83    | 268   | 153   | 35    | 2     | 0     | 0     |
| Wind                                                                             | WSAvg       | 5.1    | 5.1   | 5.0   | 5.5   | 6.0   | 6.0   | 4.9   | 4.2   | 4.1   | 5.0   | 5.4   | 5.4   | 5.0   |
| Precipitation                                                                    | PrecAvg     | 27.7   | 0.8   | 0.6   | 1.1   | 2.0   | 3.0   | 4.1   | 4.3   | 3.3   | 3.3   | 2.8   | 1.5   | 1.0   |
|                                                                                  | PrecMax     | 35.6   | 2.0   | 1.4   | 3.3   | 4.9   | 6.3   | 8.0   | 10.0  | 5.6   | 7.7   | 5.5   | 3.3   | 1.6   |
|                                                                                  | PrecMin     | 21.3   | 0.2   | 0.0   | 0.2   | 0.9   | 1.1   | 1.1   | 2.0   | 1.0   | 0.9   | 0.2   | 0.0   | 0.2   |
|                                                                                  | PrecStd     | 3.4    | 0.5   | 0.4   | 0.7   | 1.0   | 1.3   | 1.5   | 2.0   | 1.4   | 1.4   | 1.3   | 0.9   | 0.4   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 43.0  | 45.9  | 64.4  | 73.5  | 83.9  | 86.8  | 89.7  | 89.6  | 84.0  | 75.5  | 62.7  | 42.7  |
|                                                                                  |             | MCWB   | 36.8  | 39.7  | 53.1  | 57.1  | 65.1  | 72.6  | 72.6  | 73.6  | 68.8  | 58.8  | 53.2  | 37.7  |
|                                                                                  | 2%          | DB     | 36.2  | 39.4  | 55.4  | 68.3  | 79.0  | 81.8  | 85.8  | 83.8  | 80.5  | 69.6  | 54.9  | 37.3  |
|                                                                                  |             | MCWB   | 33.2  | 35.4  | 46.7  | 51.6  | 60.6  | 67.4  | 71.6  | 70.4  | 67.1  | 57.3  | 48.7  | 34.7  |
|                                                                                  | 5%          | DB     | 32.4  | 36.2  | 49.8  | 63.6  | 73.3  | 79.2  | 82.3  | 81.1  | 75.0  | 63.4  | 49.8  | 35.6  |
|                                                                                  |             | MCWB   | 30.8  | 32.3  | 43.5  | 48.7  | 57.5  | 65.1  | 69.6  | 68.2  | 64.5  | 54.1  | 44.1  | 33.2  |
|                                                                                  | 10%         | DB     | 28.3  | 31.7  | 45.0  | 58.6  | 70.0  | 75.1  | 80.9  | 77.4  | 72.1  | 57.4  | 45.2  | 32.2  |
|                                                                                  |             | MCWB   | 26.8  | 29.7  | 39.0  | 45.9  | 56.2  | 63.1  | 68.5  | 65.4  | 62.7  | 49.8  | 40.9  | 31.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 37.5  | 42.1  | 56.8  | 59.4  | 68.5  | 75.4  | 78.1  | 76.4  | 73.5  | 62.5  | 54.6  | 38.4  |
|                                                                                  |             | MCDB   | 41.9  | 45.8  | 62.2  | 72.1  | 77.8  | 83.6  | 83.9  | 83.3  | 79.8  | 68.4  | 58.3  | 40.2  |
|                                                                                  | 2%          | WB     | 33.4  | 36.0  | 48.1  | 53.4  | 64.3  | 71.2  | 74.9  | 72.8  | 69.3  | 58.7  | 49.1  | 35.6  |
|                                                                                  |             | MCDB   | 35.4  | 40.1  | 55.4  | 63.1  | 73.2  | 78.7  | 81.5  | 81.1  | 76.1  | 68.1  | 56.5  | 38.0  |
|                                                                                  | 5%          | WB     | 30.5  | 32.3  | 42.8  | 50.4  | 60.9  | 68.2  | 72.4  | 70.3  | 66.7  | 55.0  | 44.9  | 33.0  |
|                                                                                  |             | MCDB   | 32.1  | 35.5  | 49.3  | 61.7  | 69.1  | 75.7  | 79.8  | 77.9  | 73.0  | 62.0  | 48.9  | 34.6  |
|                                                                                  | 10%         | WB     | 27.4  | 29.0  | 38.8  | 46.7  | 58.0  | 65.7  | 70.1  | 68.0  | 63.8  | 51.5  | 40.7  | 30.5  |
|                                                                                  |             | MCDB   | 28.4  | 31.1  | 44.2  | 57.1  | 66.6  | 72.6  | 78.2  | 74.9  | 69.6  | 57.1  | 45.2  | 31.8  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 20.7  | 22.9  | 22.2  | 22.3  | 24.3  | 23.1  | 24.0  | 24.3  | 22.6  | 18.6  | 15.7  | 16.6  |
|                                                                                  |             | MCDBR  | 18.0  | 22.4  | 25.6  | 32.1  | 33.1  | 26.8  | 26.4  | 27.5  | 26.5  | 26.7  | 22.7  | 16.5  |
|                                                                                  |             | MCWBR  | 15.5  | 17.2  | 16.9  | 18.5  | 17.5  | 14.8  | 14.6  | 15.4  | 15.6  | 16.2  | 15.3  | 13.9  |
|                                                                                  | 5% WB       | MCDBR  | 17.3  | 19.3  | 23.4  | 28.1  | 25.5  | 21.8  | 22.8  | 22.9  | 22.0  | 23.2  | 20.7  | 15.7  |
|                                                                                  |             | MCWBR  | 15.2  | 15.0  | 16.2  | 17.5  | 15.0  | 14.0  | 14.4  | 14.1  | 14.5  | 14.9  | 15.1  | 14.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.251 | 0.270 | 0.307 | 0.346 | 0.358 | 0.369 | 0.382 | 0.372 | 0.348 | 0.306 | 0.282 | 0.261 |
|                                                                                  | taud        |        | 2.381 | 2.335 | 2.385 | 2.380 | 2.398 | 2.389 | 2.375 | 2.429 | 2.476 | 2.581 | 2.524 | 2.407 |
|                                                                                  | Ebn at Noon |        | 269   | 286   | 291   | 287   | 287   | 283   | 278   | 276   | 273   | 272   | 256   | 250   |
|                                                                                  | Edh at Noon |        | 21    | 28    | 32    | 35    | 36    | 36    | 37    | 33    | 29    | 22    | 19    | 18    |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 445   | 812   | 1185  | 1491  | 1670  | 1781  | 1906  | 1612  | 1193  | 708   | 405   | 315   |
|                                                                                  | RadStd      |        | 53    | 77    | 113   | 144   | 115   | 136   | 92    | 119   | 93    | 99    | 51    | 33    |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (38 neighbors) |       | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page